

# Route G-R4: A Physical Source and the Cross-Transition Universality Test

Thermal electric-dipole response, without a fitted time law

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## Abstract

We select one physical source whose coupling is known: an isotropic blackbody radiation bath acting on two clock transitions of the same  $^{171}\text{Yb}^+$  ion. Independently measured polarizabilities fix both frequency-shift signs. The leading fractional responses differ by a factor of about 7.25, excluding a sole common rescaling as a description of this selected thermal channel. An additional exactly common multiplicative factor is invisible to the same-bath frequency ratio and is not excluded. This is a conventional electromagnetic control, not a derivation of emergent time or an identification of radiation density with the source document's  $E_d$ .

## 1 One declared source, two independently calibrated responses

Retain the G-R3 lesson: a density observable does not fix a coupling or its sign. Here the source is thermal electromagnetic radiation, the clock is one ion, and the microscopic interaction is

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}(\mathbf{x}_{\text{ion}}). \quad (1)$$

The two transitions share the  $^2S_{1/2}(F=0)$  ground level  $g$ : E2 ends at  $^2D_{3/2}(F=2)$  and E3 at  $^2F_{7/2}(F=3)$ . The labels E2/E3 describe the *clock interrogation* multipoles; the selected thermal shift is an *electric-dipole* response of their levels. An isotropic bath removes the tensor contribution.

Use the single sign convention  $\Delta\alpha_i = \alpha_{e_i} - \alpha_g$ . The independent applied-field determinations used here are

$$\Delta\alpha_2(0) = 6.9(1.4) 10^{-40}, \quad \Delta\alpha_3(0) = 0.888(0.016) 10^{-40} \quad [\text{J m}^2/\text{V}^2]. \quad (2)$$

The E2 source [1] reports the *opposite* convention,  $\alpha_g - \alpha_{e_2} = -6.9(1.4) 10^{-40}$ ; E3 [2] uses excited minus ground. These are not coefficients adjusted to our proposed temperature comparison, nor claimed to be the latest possible calibration. Their uncertainties and provenance are retained.

**Scope boundary.** This realization assumes ordinary space, QED and weak-field atomic spectroscopy. Its  $u_\gamma$  is thermal energy per *physical volume*, not G-R3 excitation energy per addressed cell and not an operational definition of Chen's  $E_d$ . It does not complete G-R2's recording apparatus or replace the finite G-R3 stationary model. It tests a proposed physical interpretation.

## 2 Derive the thermal response and its sign

For a monochromatic field with peak amplitude  $E_0$  and polarization  $\epsilon$ , second-order perturbation theory gives

$$\delta E_a(\omega) = \frac{E_0^2}{4} \sum_b |\langle b | \mathbf{d} \cdot \epsilon | a \rangle|^2 \left[ \frac{1}{E_a - E_b + \hbar\omega} + \frac{1}{E_a - E_b - \hbar\omega} \right] \quad (3)$$

$$= -\frac{1}{4} \alpha_a(\omega) E_0^2 = -\frac{1}{2} \alpha_a(\omega) \langle E^2 \rangle. \quad (4)$$

Principal values select the dispersive real shift; resonant excitation, linewidths and scattering are not included in this expression. After isotropic polarization averaging, a fixed-state scalar response is

$$\alpha_a^{\text{sc}}(\omega) = \frac{2}{3} \sum_b \frac{(E_b - E_a) \sum_{j=x,y,z} |\langle b | d_j | a \rangle|^2}{(E_b - E_a)^2 - (\hbar\omega)^2}. \quad (5)$$

The sum includes the relevant substates. Upward and downward virtual transitions contribute with different signs. Even if each individual level polarizability were positive, its *difference* need not be. Thus positivity of energy density alone cannot fix a clock-shift sign. Equation (2) supplies the independently measured sign.

For the two-polarization Planck distribution, excluding vacuum energy,

$$u_\omega(T) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/(k_B T)} - 1}, \quad u_\gamma = \int_0^\infty u_\omega d\omega = a_{\text{rad}} T^4, \quad (6)$$

$$a_{\text{rad}} = \frac{8\pi^5 k_B^4}{15h^3 c^3}, \quad \langle E^2 \rangle = \frac{u_\gamma}{\epsilon_0}. \quad (7)$$

The integral follows from  $\int_0^\infty x^3/(e^x - 1) dx = \pi^4/15$ . Electric and magnetic fields each carry half the thermal energy; using  $2u_\gamma/\epsilon_0$  for the mean squared electric field would double-count it. Combining the level shifts yields

$$\delta\nu_i(T) = -\frac{1}{2h\epsilon_0} \mathcal{P} \int_0^\infty \Delta\alpha_i(\omega) u_\omega(T) d\omega, \quad (8)$$

$$s_i(T) := \frac{\delta\nu_i}{\nu_i^0} = \beta_i u_\gamma [1 + \eta_i(T)], \quad \beta_i = -\frac{\Delta\alpha_i(0)}{2h\epsilon_0\nu_i^0}. \quad (9)$$

Here  $1 + \eta_i$  is the spectral integral divided by  $\Delta\alpha_i(0)u_\gamma$ , not a fitted slowdown function. The leading static approximation sets  $\eta_i = 0$ . Positive differential polarizabilities give negative shifts for both chosen transitions. This is the conventional BBR shift formula, also used in the 2024 clock analysis [3].

### 3 What equal fractional shifts would actually require

**Diagonal-level criterion.** Let  $H_0$  have level energies  $E_a$  and let  $\delta E_a$  be their secular first-order shifts. On a connected graph of measured transitions, every edge has the same fractional shift  $f$  if and only if

$$\delta E_a = f E_a + C \quad \text{on those levels,} \quad \delta H_{\text{sec}} = f H_0 + C \mathbb{I}. \quad (10)$$

*Proof.* On an edge  $(a, b)$ , the hypothesis is  $\delta E_a - \delta E_b = f(E_a - E_b)$ , so  $\delta E_a - f E_a = \delta E_b - f E_b$ . Connectivity makes this value one constant  $C$ . Conversely that form gives the stated shift on every edge. This is a statement about diagonal energy shifts, not an operator identity for arbitrary off-diagonal perturbations or a proof of universality for untested transitions.

For the three levels  $g, e_2, e_3$ , the selected leading thermal response would therefore require

$$\frac{\Delta\alpha_2(0)}{\nu_2^0} = \frac{\Delta\alpha_3(0)}{\nu_3^0}, \quad \beta_2 = \beta_3. \quad (11)$$

The independent inputs violate this condition. In the off-resonant low-temperature expansion, analytic  $\Delta\alpha_i(\omega)$  gives  $s_i = \beta_i a_{\text{rad}} T^4 + O(T^6)$ . Different leading coefficients cannot be made identical on a low-temperature interval by  $T^6$  terms. This asymptotic observation is not a full finite-temperature error bound or an exclusion of accidental equality at one temperature.

Define  $R(u) = \nu_3(u)/\nu_2(u)$  and  $R_0 = \nu_3^0/\nu_2^0$ . Within the retained static-response model, exact quotient algebra gives

$$\frac{R(u)}{R_0} - 1 = \frac{(\beta_3 - \beta_2)u}{1 + \beta_2 u}, \quad (12)$$

$$\frac{R(u_h)}{R(u_l)} - 1 = \frac{(\beta_3 - \beta_2)(u_h - u_l)}{(1 + \beta_2 u_h)(1 + \beta_3 u_l)}. \quad (13)$$

Keeping these denominators is convenient algebra, not control of all higher orders in the microscopic interaction. The expressions avoid subtracting nearly equal floating-point numbers.

**Identifiability limit.** Suppose an additional common physical factor exists:

$$\nu_i(u) = F(u)\nu_i^0[1 + s_i(u)], \quad R(u) = R_0 \frac{1 + s_3(u)}{1 + s_2(u)}. \quad (14)$$

$F$  cancels exactly, even when it changes between setpoints. A nonconstant ratio rejects a *sole* common factor as the explanation of this thermal response; it does not exclude a common factor in addition to it. Equal fractions would be necessary, not sufficient, evidence for a universal time effect. Relative phases obey  $d\Phi_3/d\Phi_2 = \nu_3/\nu_2$  in a quasistatic environment, without needing a preferred external time unit for that comparison.

## 4 Pinned numerical prediction and its uncertainty boundary

Use rounded frequencies  $\nu_2^0 = 688.359$  THz and  $\nu_3^0 = 642.121$  THz, exact SI  $h, k_B, c$ , and rounded  $\epsilon_0 = 8.8541878 \cdot 10^{-12}$  F/m. These suffice for a response-structure test, not a metrological absolute-frequency recommendation. At  $T = 300$  K,

$$u_\gamma = 6.12824 \cdot 10^{-6} \text{ J/m}^3, \quad E_{\text{rms}} = 831.943 \text{ V/m}. \quad (15)$$

The static electric-dipole predictions ( $\eta_2 = \eta_3 = 0$ ) are:

| Transition | $\delta\nu$ [Hz] | $s = \delta\nu/\nu^0$   |
|------------|------------------|-------------------------|
| E2         | -0.3604          | $-5.235 \cdot 10^{-16}$ |
| E3         | -0.04638         | $-7.223 \cdot 10^{-17}$ |

Thus  $|\beta_2/\beta_3| \simeq 7.25$ , and

$$\frac{R(300 \text{ K})}{R(0)} - 1 \simeq 4.513 \cdot 10^{-16}, \quad \frac{R(310 \text{ K})}{R(300 \text{ K})} - 1 \simeq 6.325 \cdot 10^{-17}. \quad (16)$$

Both shifts are downward, but E2 falls by a larger fraction, so E3/E2 increases. The second number is a two-setpoint *prediction*, not an observed temperature-cycle signal. Digits document reproducibility; the polarizability uncertainty, especially E2's, sets physical precision.

The independently published E3 correction  $\eta_3(300 \text{ K}) = -0.0015(7)$  [2] changes its shift to about  $-0.04631$  Hz. We do not supply a complete E2 spectral response, full covariance or apparatus radiation model, and do not label the table as a full finite-temperature prediction.

For a transparent robustness check, allow each polarizability to vary over its separate  $\pm 3\sigma$  input interval and each dynamic factor over  $[0.9, 1.1]$ . Monotonicity gives the exact box lower bound

$$(\beta_3 - \beta_2)_{\min} = \frac{1}{2h\epsilon_0} \left[ \frac{0.9(\alpha_2 - 3\sigma_2)}{\nu_2^0} - \frac{1.1(\alpha_3 + 3\sigma_3)}{\nu_3^0} \right] = 1.642 \cdot 10^{-11} \text{ m}^3/\text{J} > 0, \quad (17)$$

where  $\alpha_i$  denotes  $\Delta\alpha_i(0)$  in this bound. The corresponding leading ratio offset at 300 K exceeds  $1.006 \cdot 10^{-16}$ . This is a *conditional deterministic stress box*, not a statistical confidence level or a derived bound on all missing dynamic corrections. The unknown cross-paper correlation  $\rho$  gives

$$\sigma_{\beta_3-\beta_2}^2 = \sigma_{\beta_3}^2 + \sigma_{\beta_2}^2 - 2\rho\sigma_{\beta_3}\sigma_{\beta_2}, \quad \sigma_{\beta_3-\beta_2} \in [|\sigma_{\beta_2} - \sigma_{\beta_3}|, \sigma_{\beta_2} + \sigma_{\beta_3}]. \quad (18)$$

No independence-based significance or new combined best value is claimed.

## 5 The next physical discriminator, not another time-law fit

**A: test the differential prediction.** Specify a real common radiation environment and infer its spectrum at the ion, not merely a remote wall temperature. Interleave E2/E3 interrogation at two stable setpoints and retain the raw ratio before subtracting the deliberately modulated BBR response. Same-ion interleaved comparisons are an established protocol [4], but no new data are analyzed here. Published BBR-corrected clock outputs are not independent temperature-response observations. Returning a flat ratio after subtracting an assumed BBR law would not validate that same law.

The temperature change may also alter tensor shifts, magnetic fields, trap motion, probe shifts and thermal gradients. These must be measured or bounded. Magnetic-dipole/quadrupole thermal contributions and linewidth effects are outside our selected scalar electric-dipole calculation; they are not asserted to vanish in a real apparatus. The sign and leading unequal responses are a completed source audit; experimental closure requires those controls, not more Gaussian scans.

**B: make a common factor observable.** If the goal is specifically  $F(u)$  in Eq. (14), provide an independently specified reference outside the changed bath, with a physical signal-comparison protocol and its own response model, or an independently predicted non-clock observable. Do not assume by definition that the reference is unaffected. A factor that changes all physical standards identically may remain a units/reparametrization freedom rather than an observable.

**C: density versus spectral content.** As a separate optional test, compare two nonthermal radiation spectra with the same integrated  $u_\gamma$  but different spectral weights. Equation (8) predicts different shifts when  $\Delta\alpha_i(\omega)$  varies. This directly tests whether scalar density is sufficient; it requires spectral calibration and is not a new free per-clock parameter.

**Completed deliverable.** G-R4 supplies a source, an independently fixed sign, the cross-transition criterion and a reproducible numerical prediction. The preferred theoretical revision is an interaction-derived *spectral response*, with a universal component treated separately and only when observable. No additional  $F(u)$  is fitted. The code’s 91 checks validate the displayed algebra and implementation, not a new empirical time effect. Input conventions, hashes, limitations and roadmap are stored alongside the TeX; earlier Route-G results are preserved.

## References

- [1] T. Schneider, E. Peik and Chr. Tamm, *Sub-Hertz Optical Frequency Comparisons between Two Trapped  $^{171}\text{Yb}^+$  Ions*, Phys. Rev. Lett. **94**, 230801 (2005), doi:10.1103/PhysRevLett.94.230801.
- [2] N. Huntemann et al., *Single-Ion Atomic Clock with  $3 \times 10^{-18}$  Systematic Uncertainty*, Phys. Rev. Lett. **116**, 063001 (2016), doi:10.1103/PhysRevLett.116.063001.
- [3] A. Tofful et al.,  *$^{171}\text{Yb}^+$  optical clock with  $2.2 \times 10^{-18}$  systematic uncertainty and absolute frequency measurements*, Metrologia **61**, 045001 (2024), doi:10.1088/1681-7575/ad53cd.
- [4] M. Filzinger et al., *Improved limits on the coupling of ultralight bosonic dark matter to photons from optical atomic clock comparisons* (2023), arXiv:2301.03433.